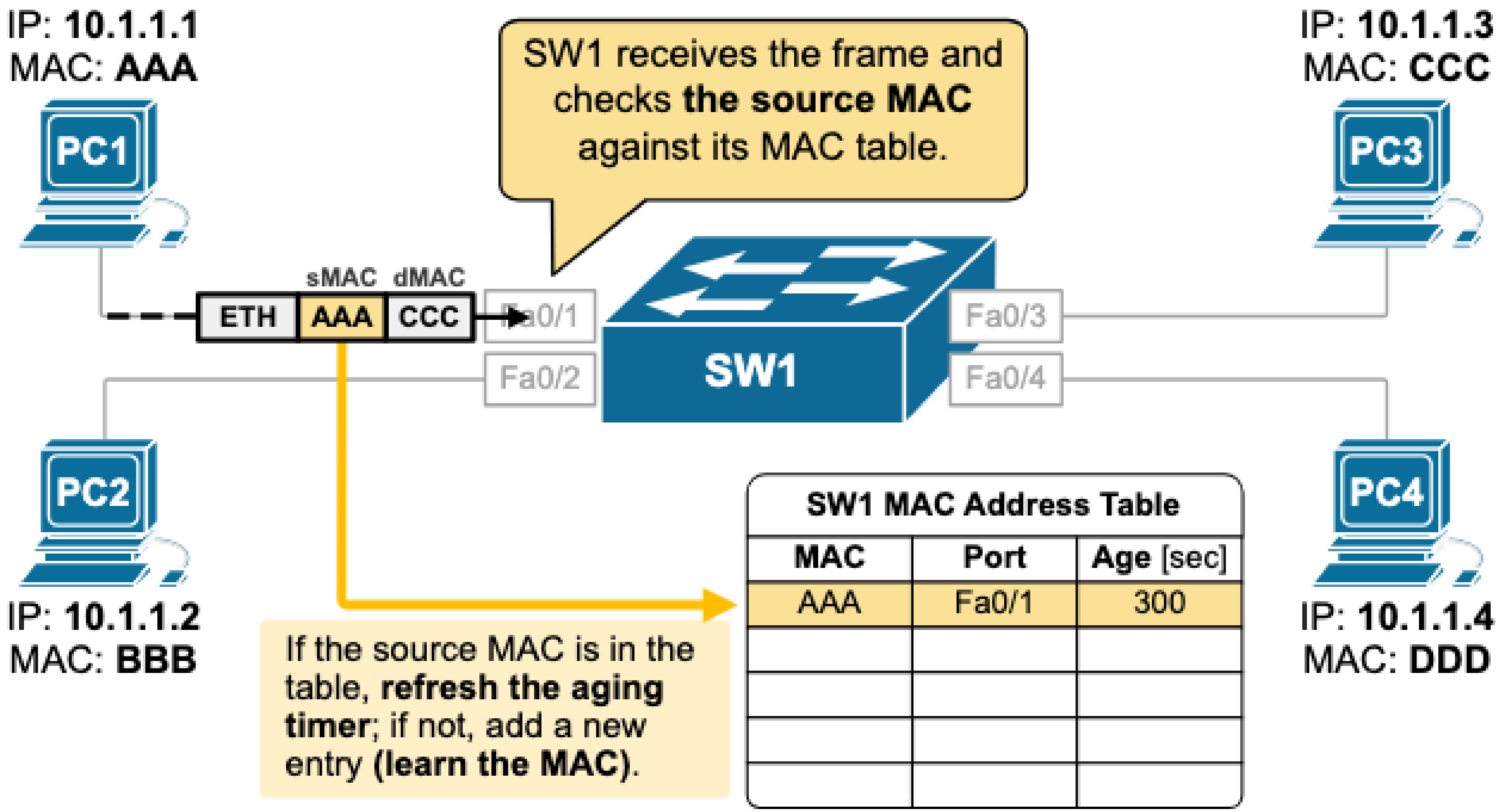
IT INFRASTRUCTURE AND COMPUTER NETWORKS

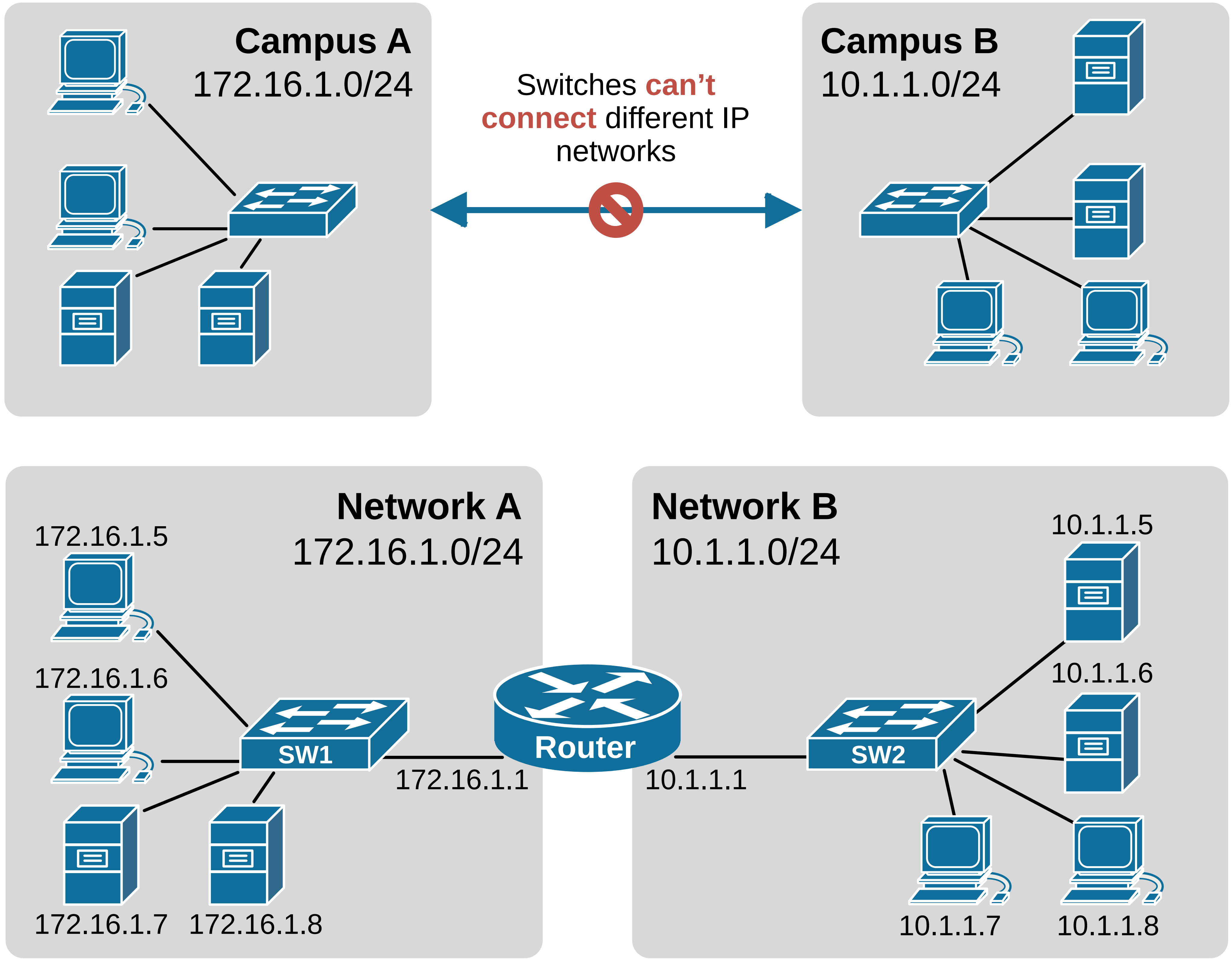
Week 8

Packets travelling within the network

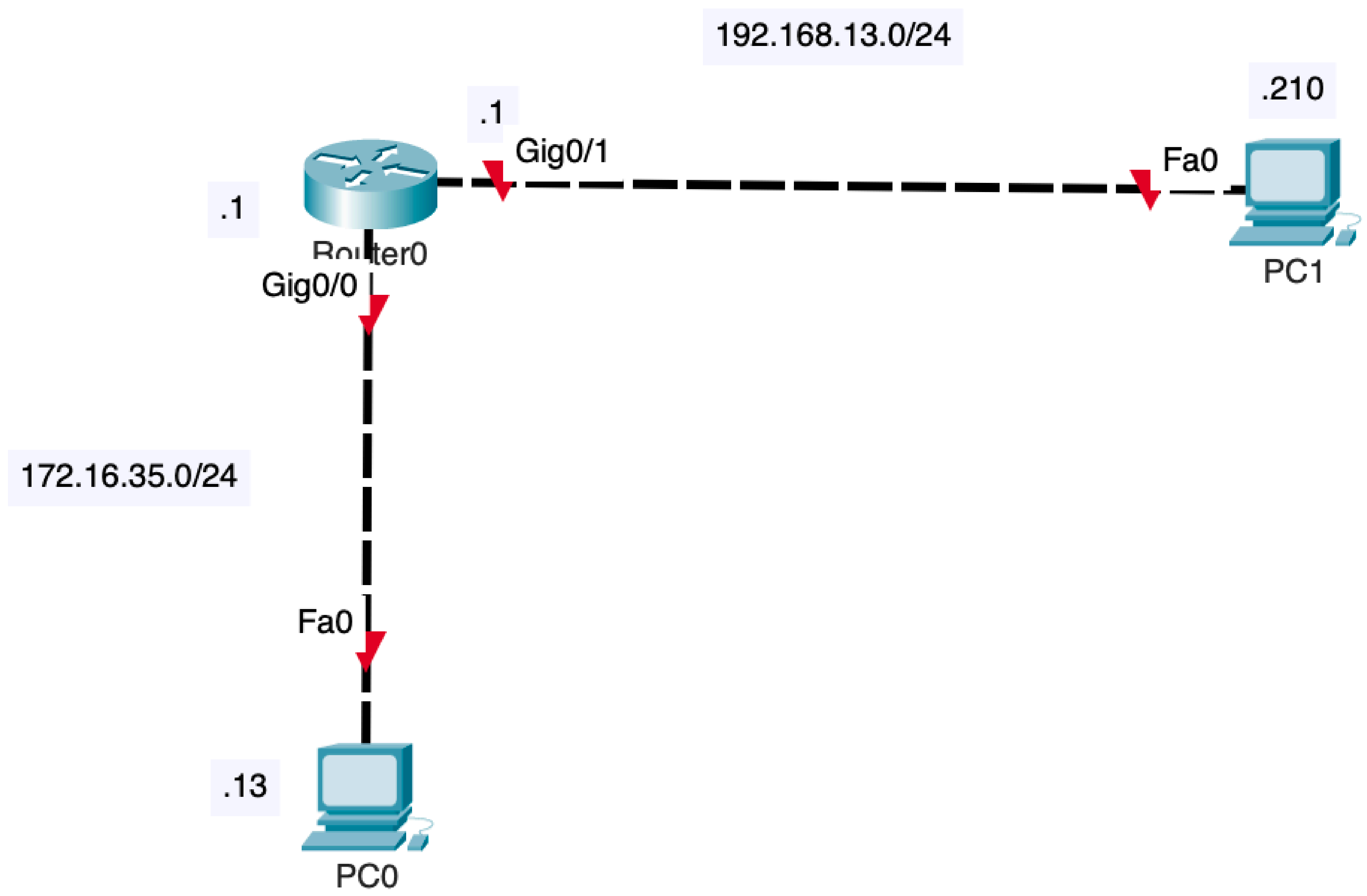
Switch is **switching packets** between source and destination devices



Problem with Switching



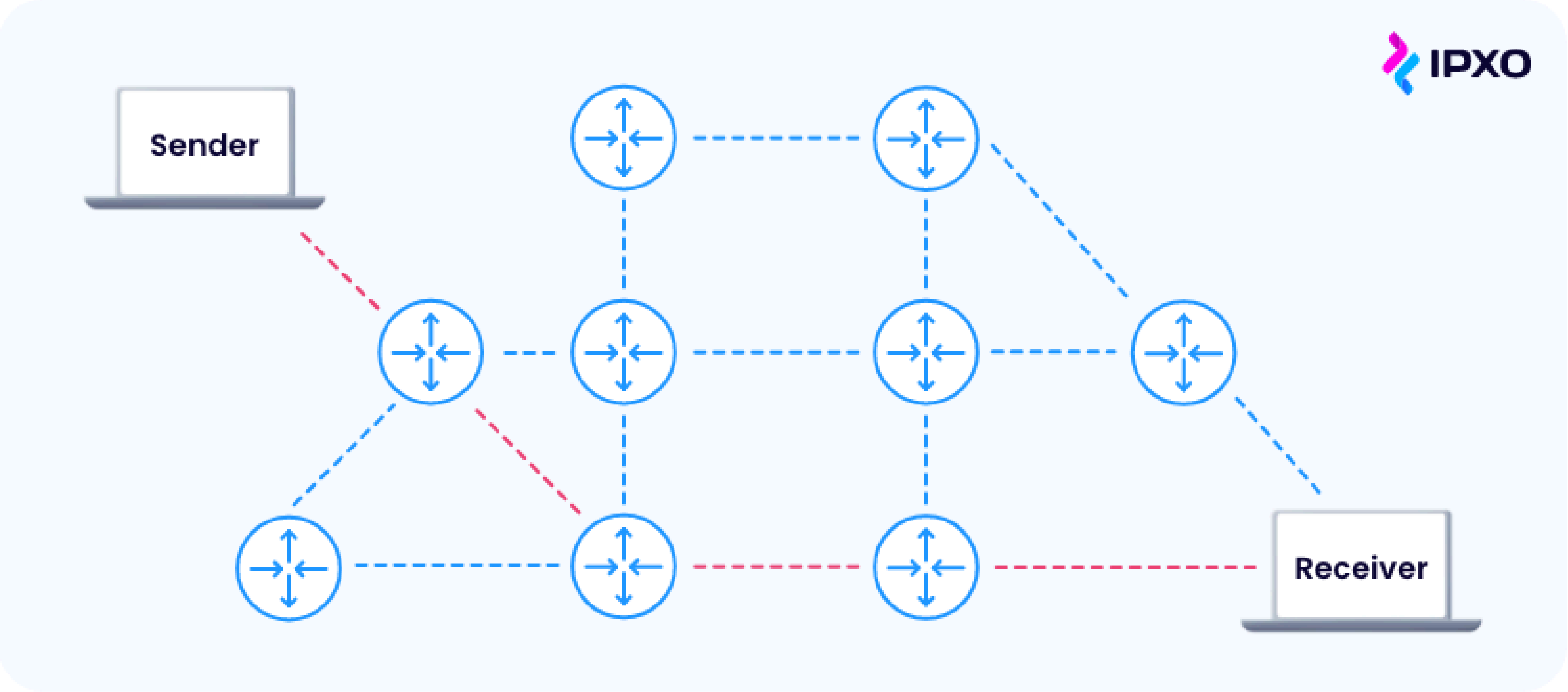
How do packets travel between different networks?



What is Routing?

Routing is the process of **selecting path** from sender to receiver

Router uses **Routing Table** to forward packets to the **Next Hop device**



How does a Router work?

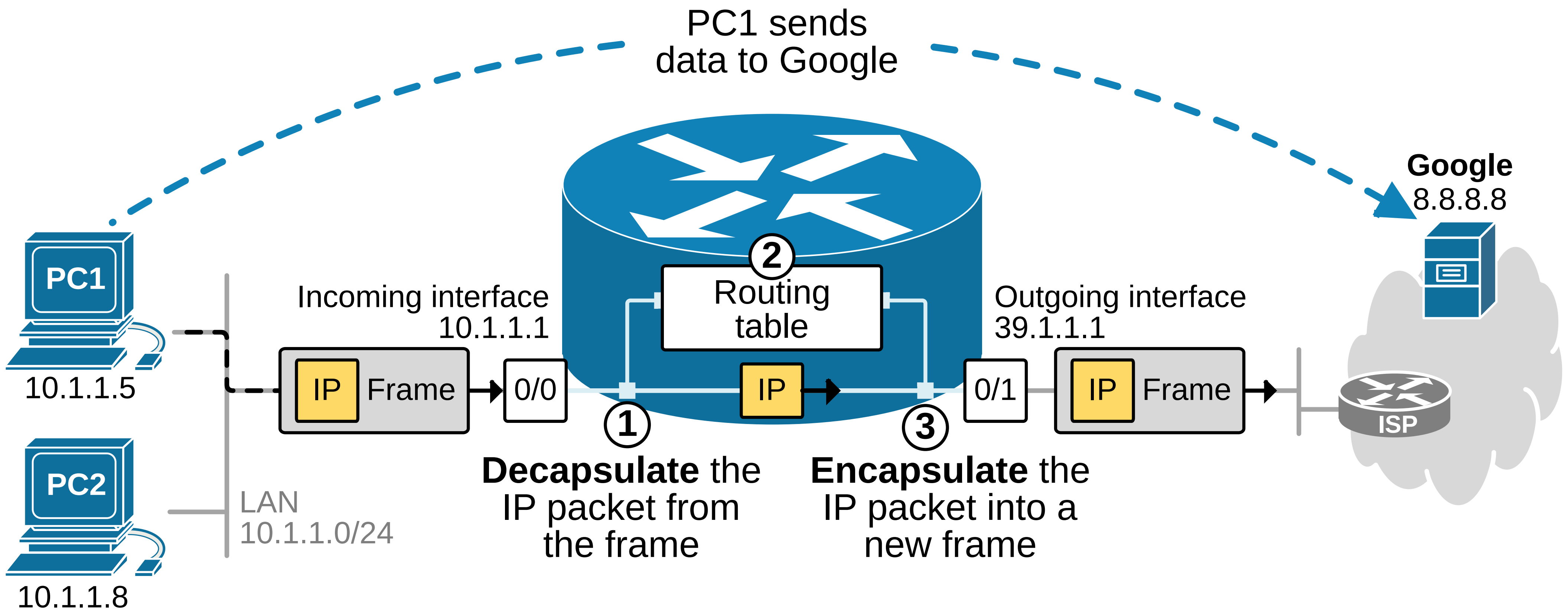
To forward packets Router uses:

1. Interfaces: Each interface is a **separate network**

   Only **one interface** can be connected to **the same network**

2. Routing Table: Tells **where to send** packets (destination network and next hop)

   Built manually and automatically



Routing Table

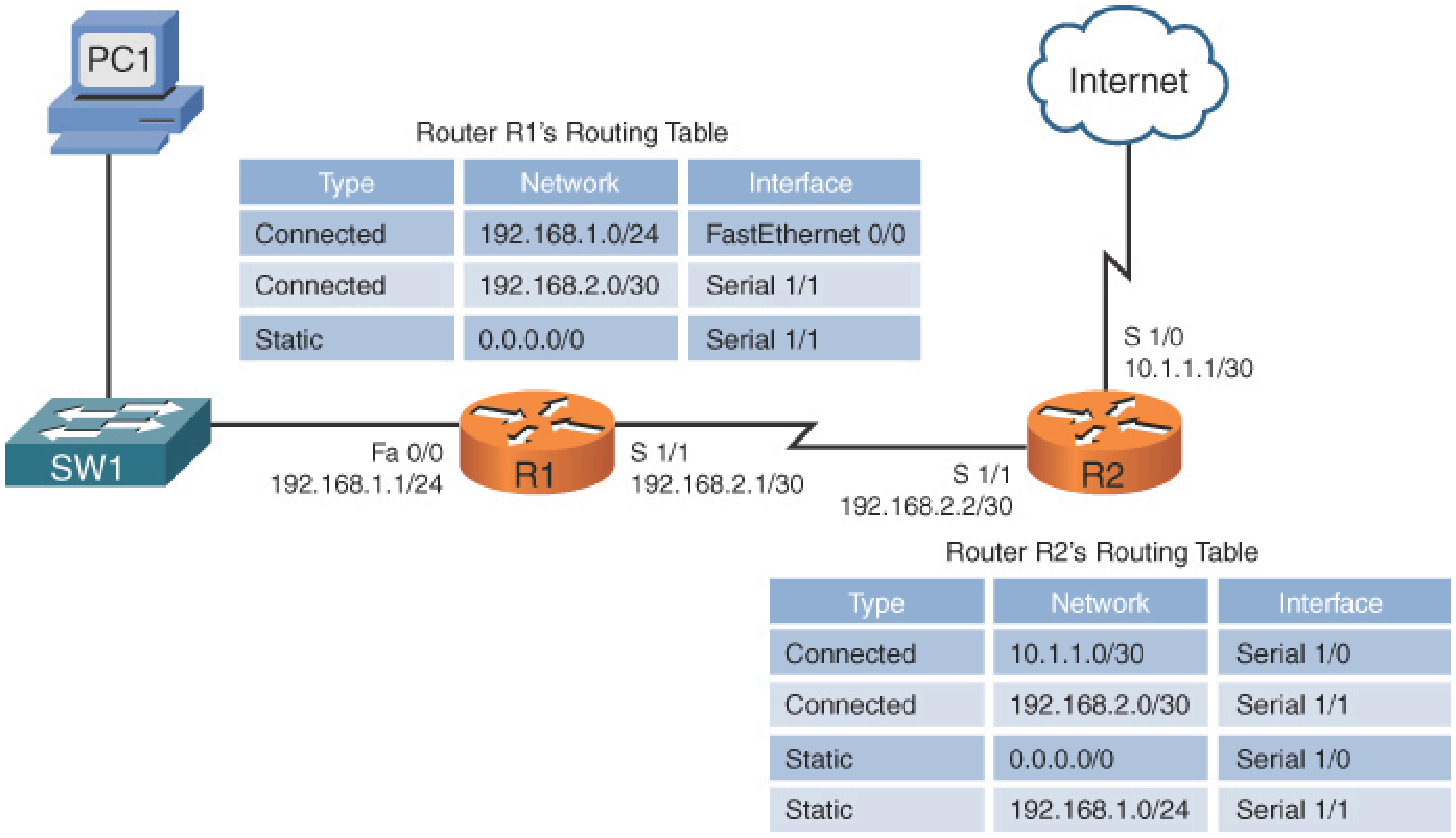
Routing table is built:

Statically (manually)

- Fixed Route
- Easy to set up, straightforward
- Requires manual updates when topology is changed
- Not adaptable

Dynamically (automatically)

- Routes are learnt through dynamic protocols
- Routes are chosen based on metrics such as distance, cost, or speed
- Adaptable in large/changing networks



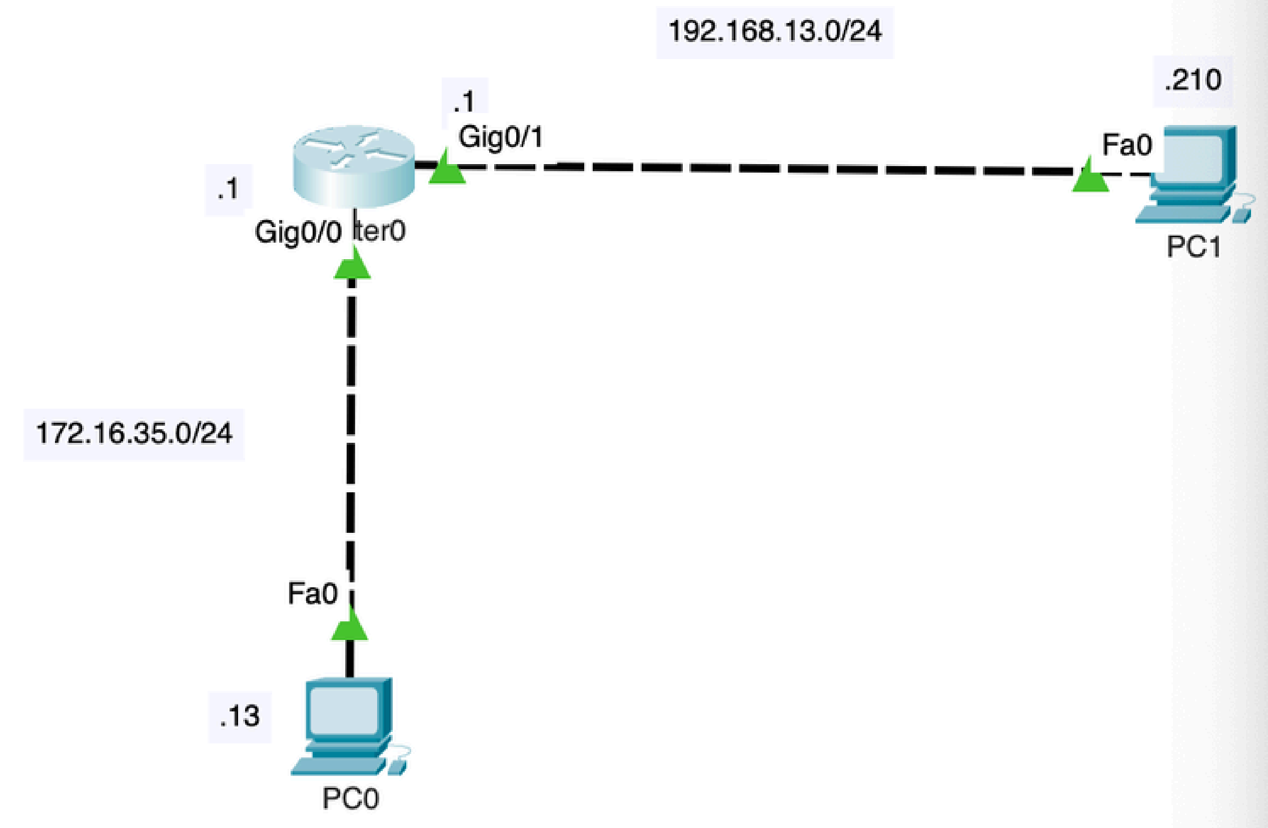
Router R1's Routing Table

Type	Network	Interface
Connected	192.168.1.0/24	FastEthernet 0/0
Connected	192.168.2.0/30	Serial 1/1
Static	0.0.0.0/0	Serial 1/1

Router R2's Routing Table

Type	Network	Interface
Connected	10.1.1.0/30	Serial 1/0
Connected	192.168.2.0/30	Serial 1/1
Static	0.0.0.0/0	Serial 1/0
Static	192.168.1.0/24	Serial 1/1

Directly connected networks and interface IP-addresses are filled in the Routing Table of the router automatically

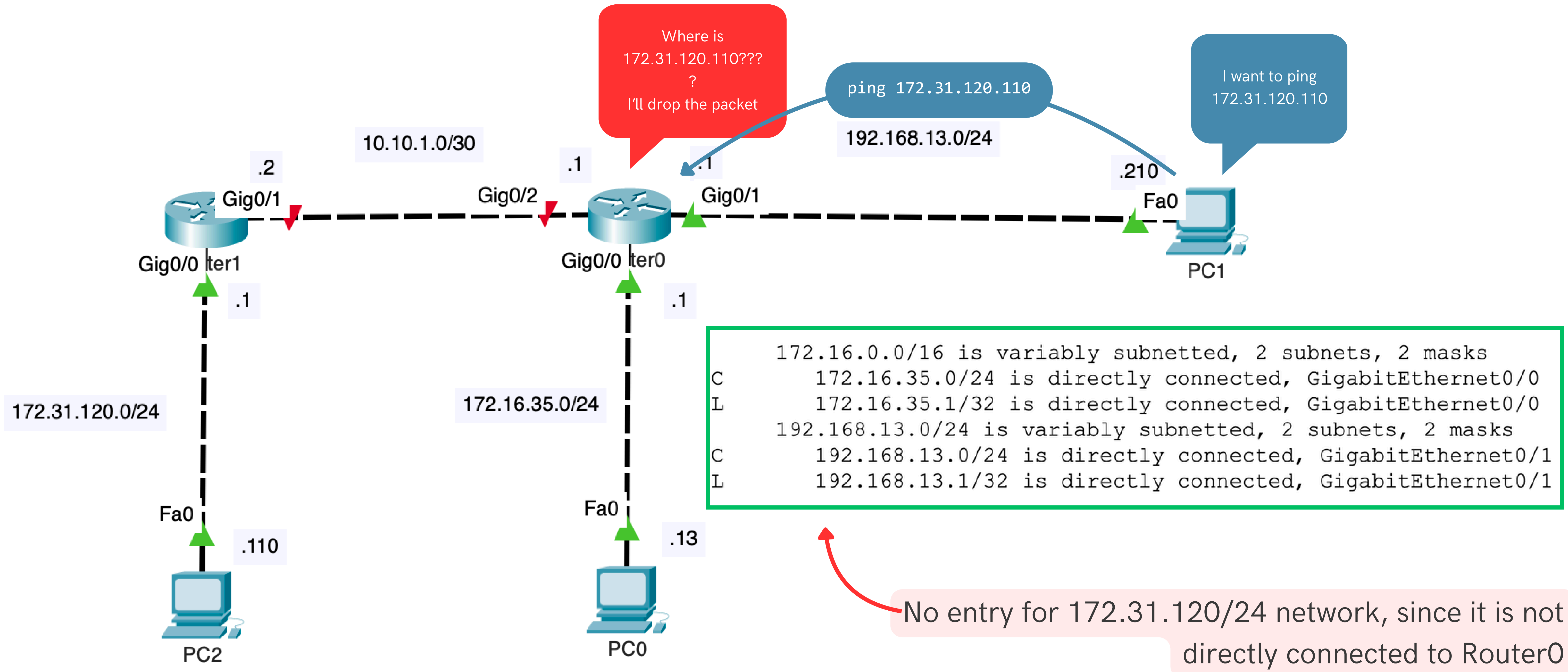



```
Router#
Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.35.0/24 is directly connected, GigabitEthernet0/0
L       172.16.35.1/32 is directly connected, GigabitEthernet0/0
     192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.13.0/24 is directly connected, GigabitEthernet0/1
L       192.168.13.1/32 is directly connected, GigabitEthernet0/1
```

C (Connected) - Directly connected network

L (Local) - IP address of Router's local interface

What about not directly connected networks?



No entry for 172.31.120/24 network, since it is not directly connected to Router0

Static Routing Configuration

To add static route to network 172.31.120.0/24 in Router0 :

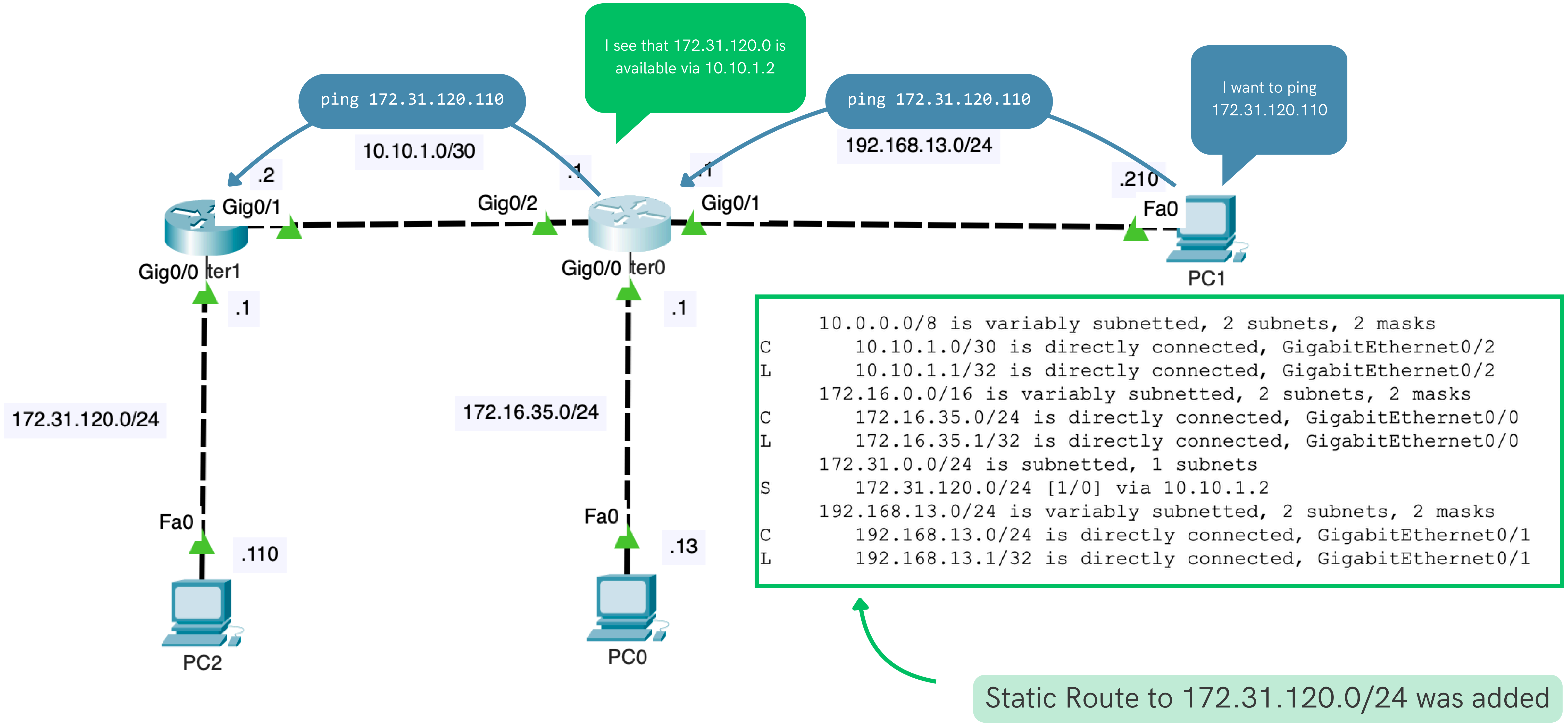
```
Router(config)#ip route 172.31.120.0 255.255.255.0 10.10.1.2
```

To see Routing Table: `Router#show ip route`

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.1.0/30 is directly connected, GigabitEthernet0/2
L       10.10.1.1/32 is directly connected, GigabitEthernet0/2
     172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.35.0/24 is directly connected, GigabitEthernet0/0
L       172.16.35.1/32 is directly connected, GigabitEthernet0/0
     172.31.0.0/24 is subnetted, 1 subnets
S       172.31.120.0/24 [1/0] via 10.10.1.2
     192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.13.0/24 is directly connected, GigabitEthernet0/1
L       192.168.13.1/32 is directly connected, GigabitEthernet0/1
```

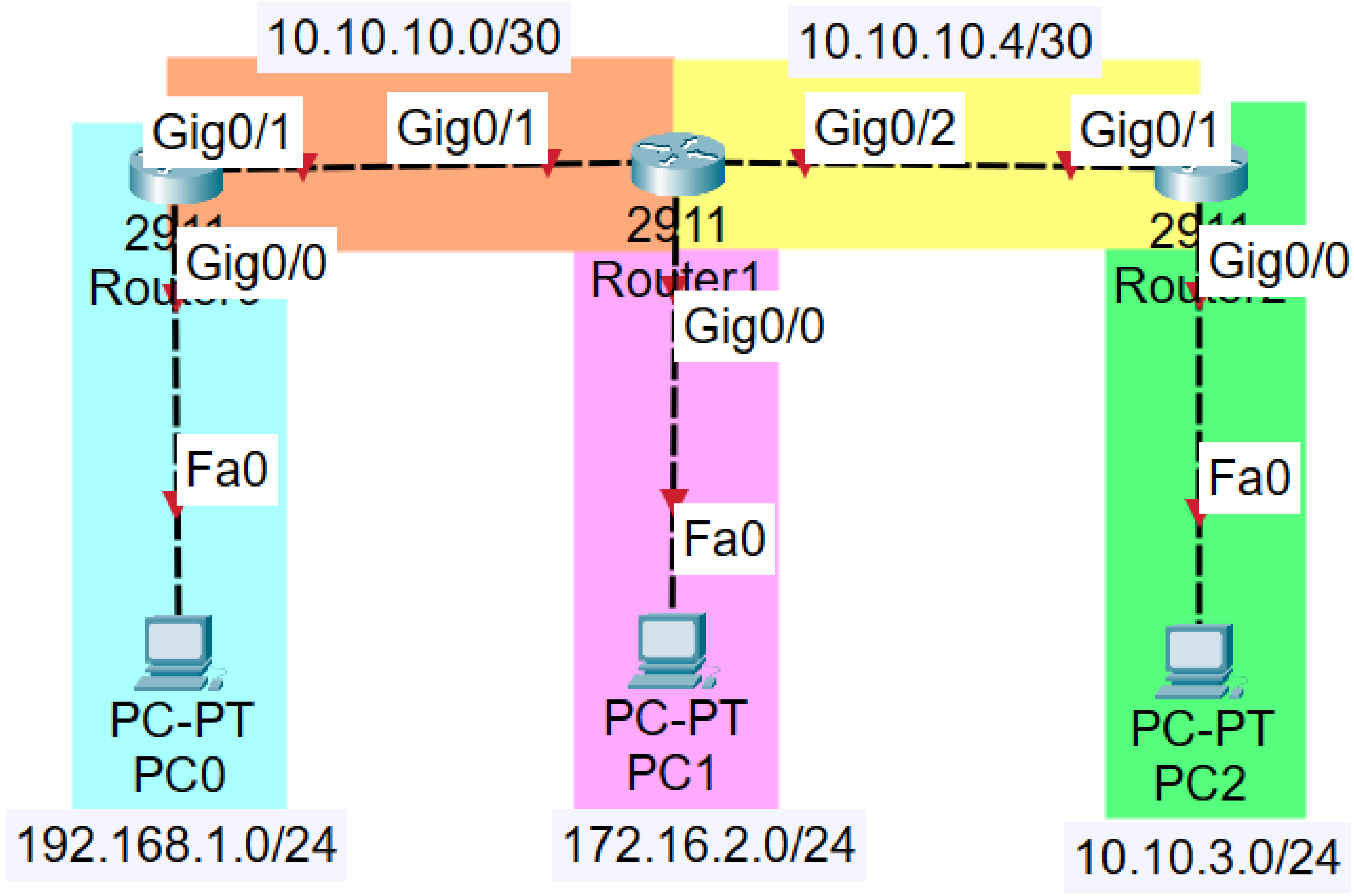
Static route is added

S (Static) - manually configured route

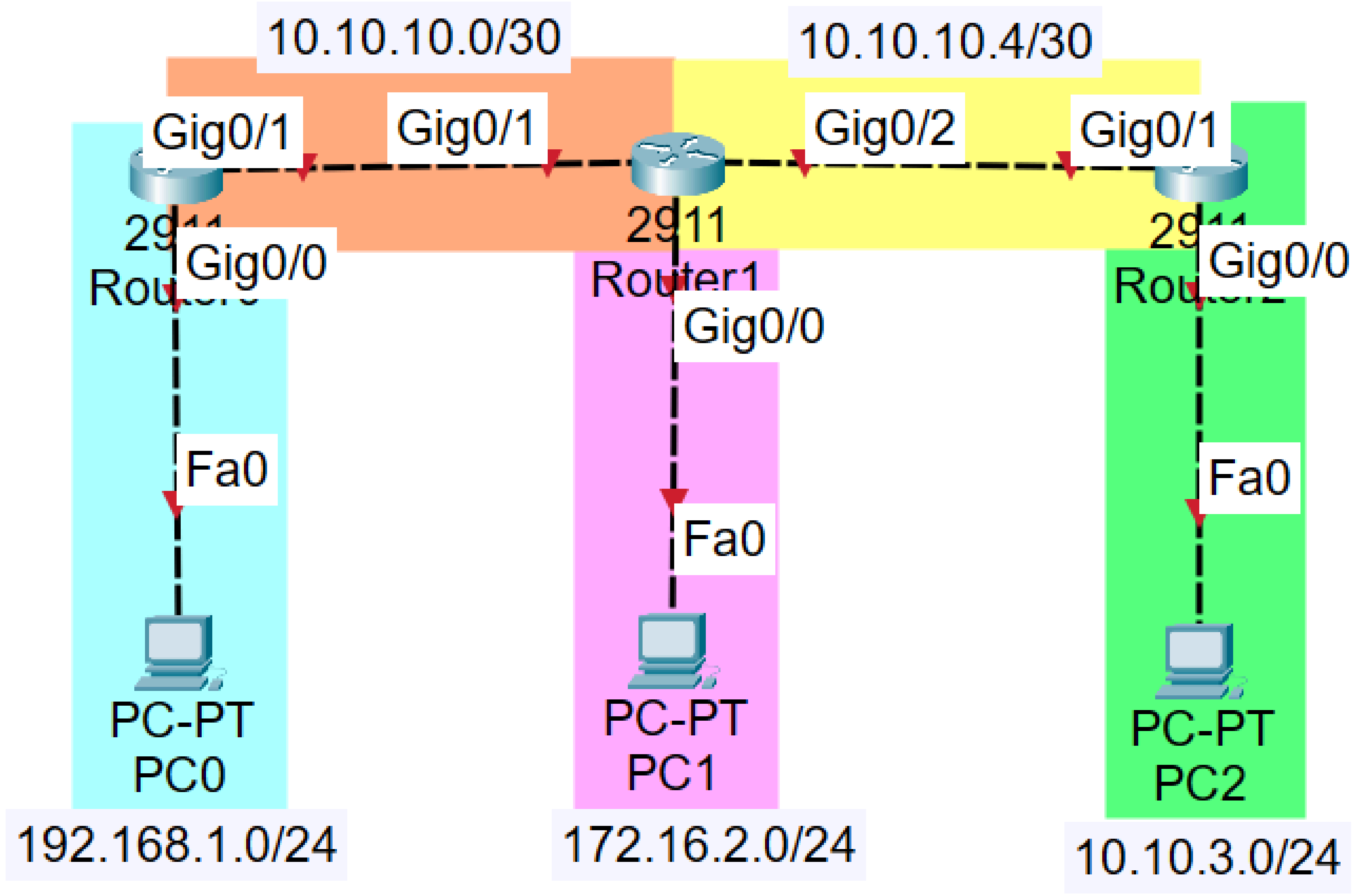


I see that 172.31.120.0 is available via 10.10.1.2
ping 172.31.120.110
ping 172.31.120.110
I want to ping 172.31.120.110
10.10.1.0/30
192.168.13.0/24
.2
.1
.1
.210
Gig0/1
Gig0/2
Gig0/1
Fa0
PC1
Gig0/0
ter1
Gig0/0
ter0
.1
.1
172.31.120.0/24
172.16.35.0/24
Fa0
.110
Fa0
.13
PC2
PC0
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.10.1.0/30 is directly connected, GigabitEthernet0/2
L 10.10.1.1/32 is directly connected, GigabitEthernet0/2
172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.16.35.0/24 is directly connected, GigabitEthernet0/0
L 172.16.35.1/32 is directly connected, GigabitEthernet0/0
172.31.0.0/24 is subnetted, 1 subnets
S 172.31.120.0/24 [1/0] via 10.10.1.2
192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.13.0/24 is directly connected, GigabitEthernet0/1
L 192.168.13.1/32 is directly connected, GigabitEthernet0/1
Static Route to 172.31.120.0/24 was added

Example of Static routing configuration



5 networks



10.10.10.0/30
10.10.10.4/30
Gig0/1
Gig0/1
Gig0/2
Gig0/1
2911
Router1
Gig0/0
Gig0/0
Gig0/0
Fa0
Fa0
Fa0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
192.168.1.0/24
172.16.2.0/24
10.10.3.0/24

Configure PCs:

PC0

192.168.1.254
255.255.255.0
192.168.1.1

PC1

172.16.2.126
255.255.255.128
172.16.2.1

PC2

10.10.3.62
255.255.255.192
10.10.3.1

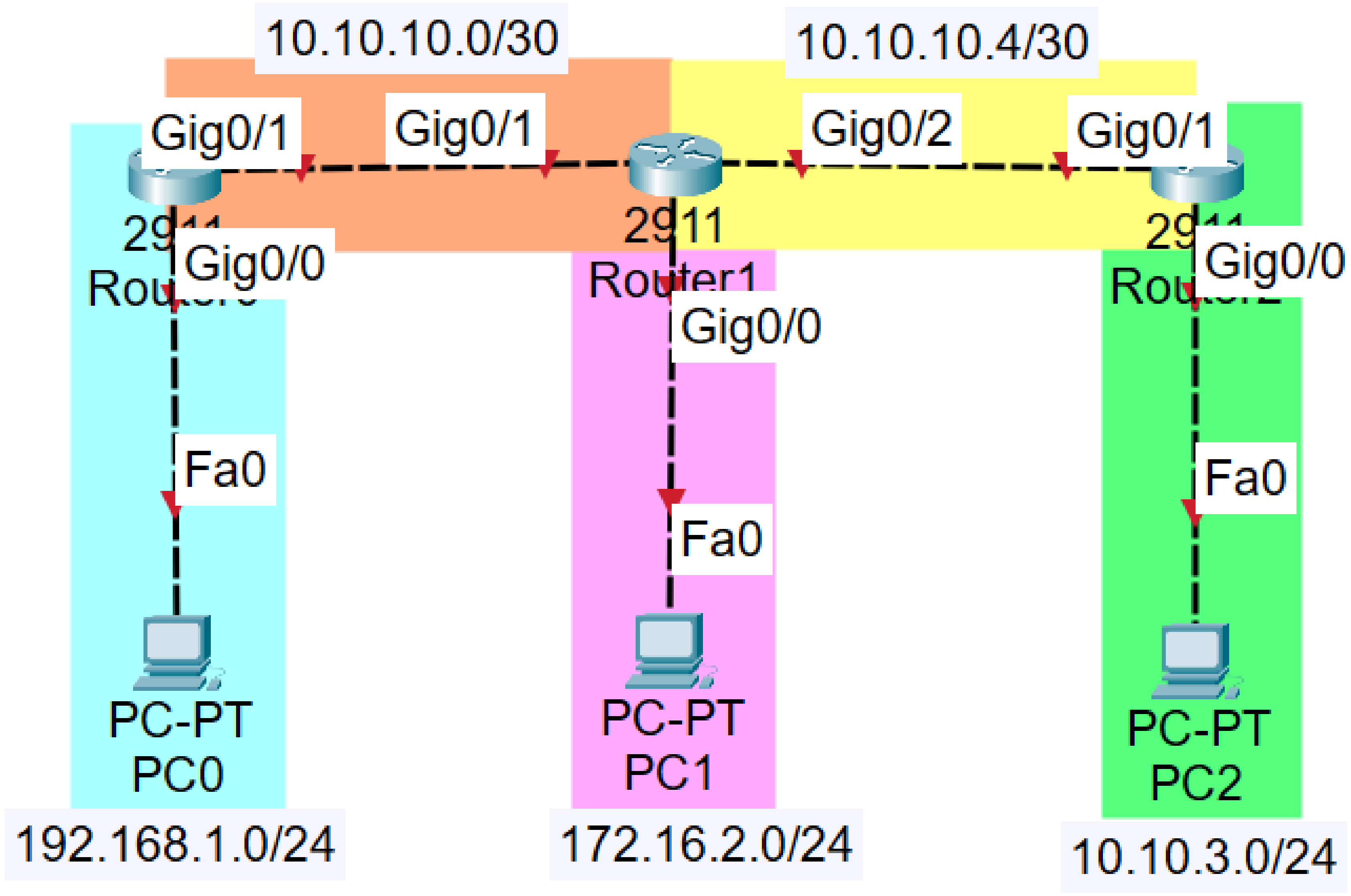
Configure Routers:

```
Router(config)#int g0/0
Router(config-if)#ip ad 192.168.1.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#int g0/1
Router(config-if)#ip ad 10.10.10.1 255.255.255.252
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
```

Assign IP address, subnet mask and default gateway for each PC
Assign IP address and subnet mask for each Router's interfaces.
Don't forget to turn on interfaces.

```
Router(config)#ip route 192.168.1.0 255.255.255.0 10.10.10.1
Router(config)#ip route 10.10.3.0 255.255.255.192 10.10.10.6
```

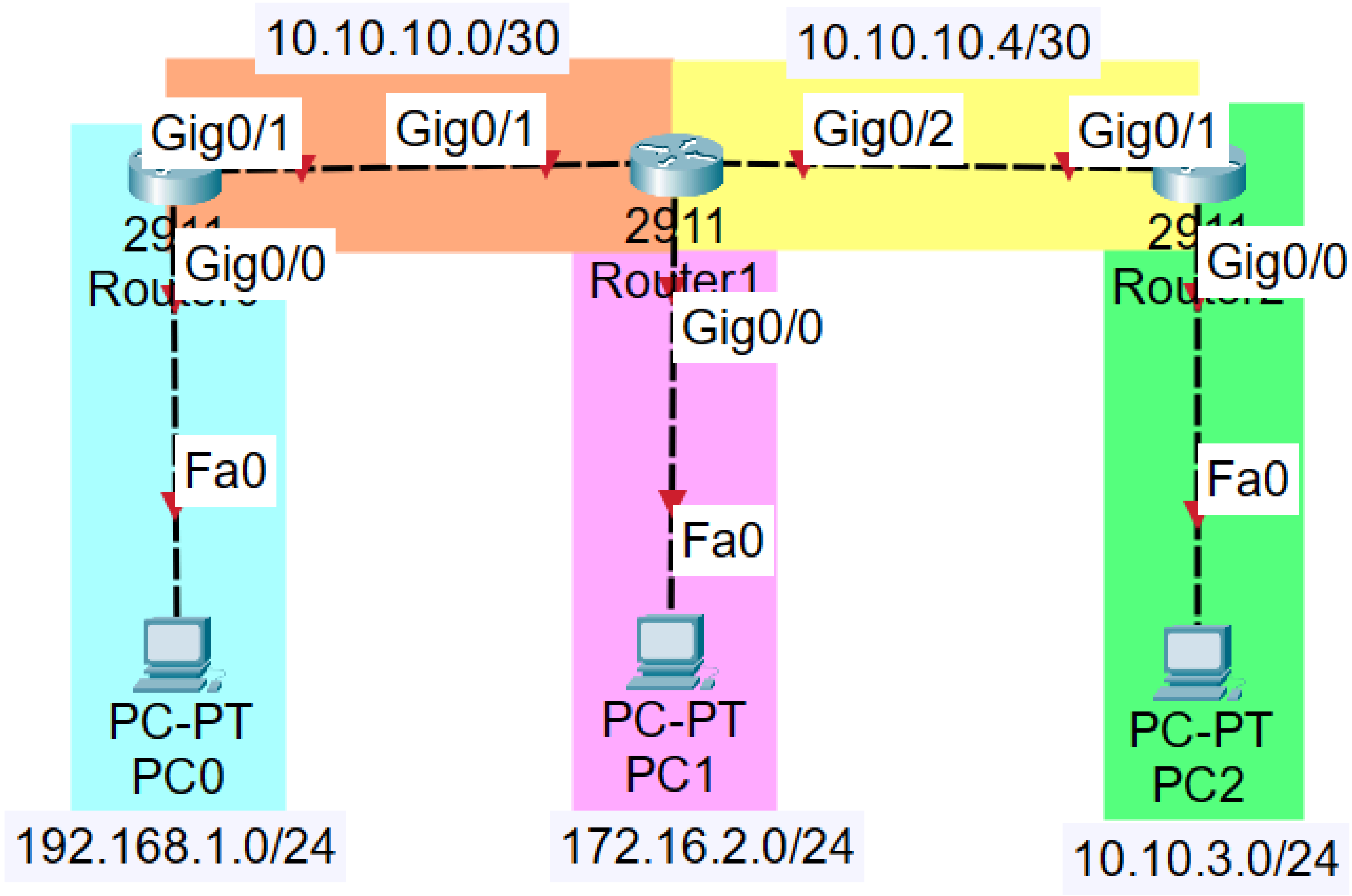
If you want to reach 192.168.1.0/24 network, ask 10.10.10.1



Router1 knows only about its direct neighbours: RT0 and RT2; and only about directly connected networks: 172.16.2.0/24, 10.10.10.0/30 and 10.10.10.4/30

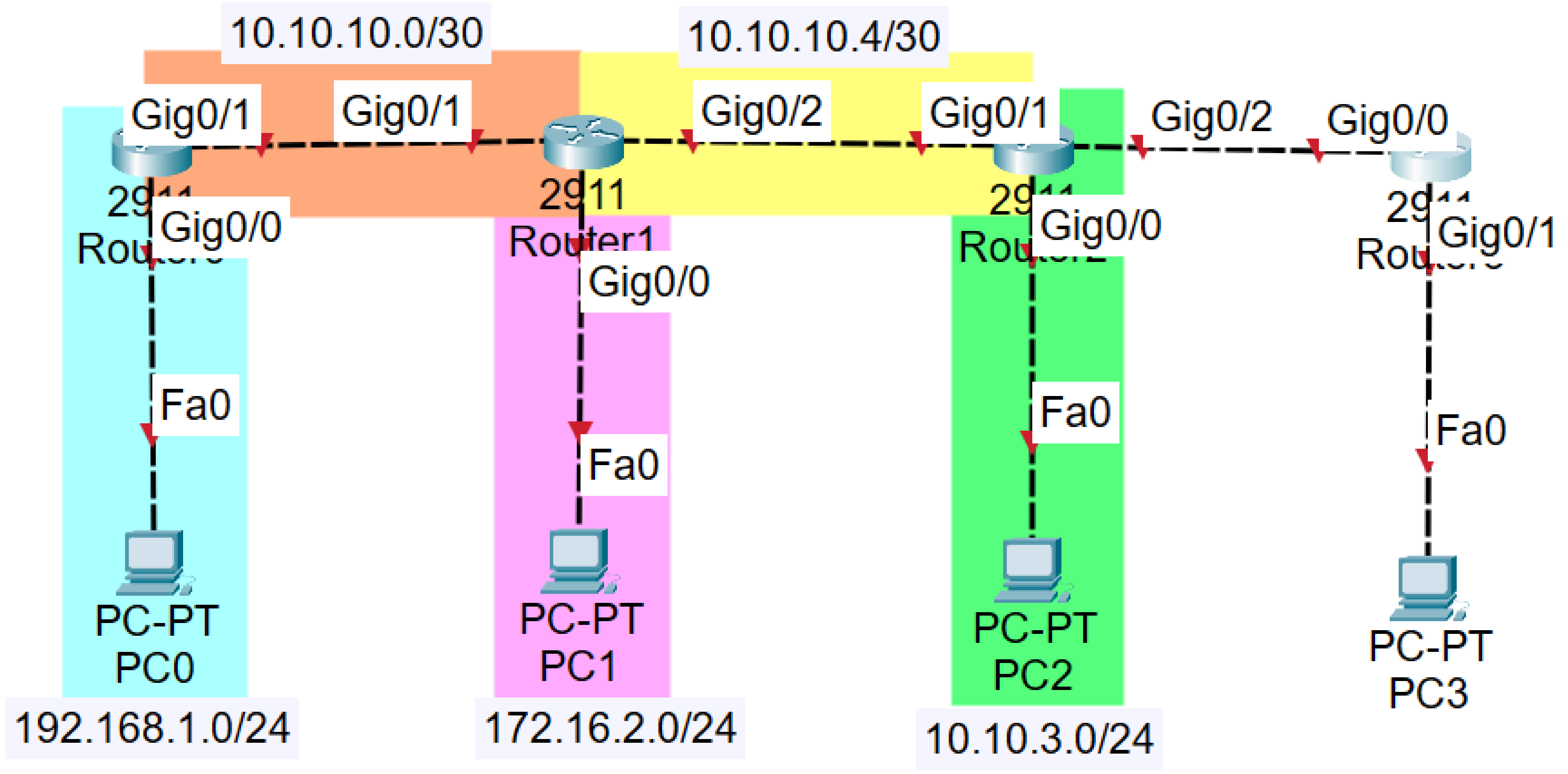
```
Router(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.2
```

If you want to reach ANY network, ask 10.10.10.2



Router0 knows only about RT1

What if we add 2 more networks?



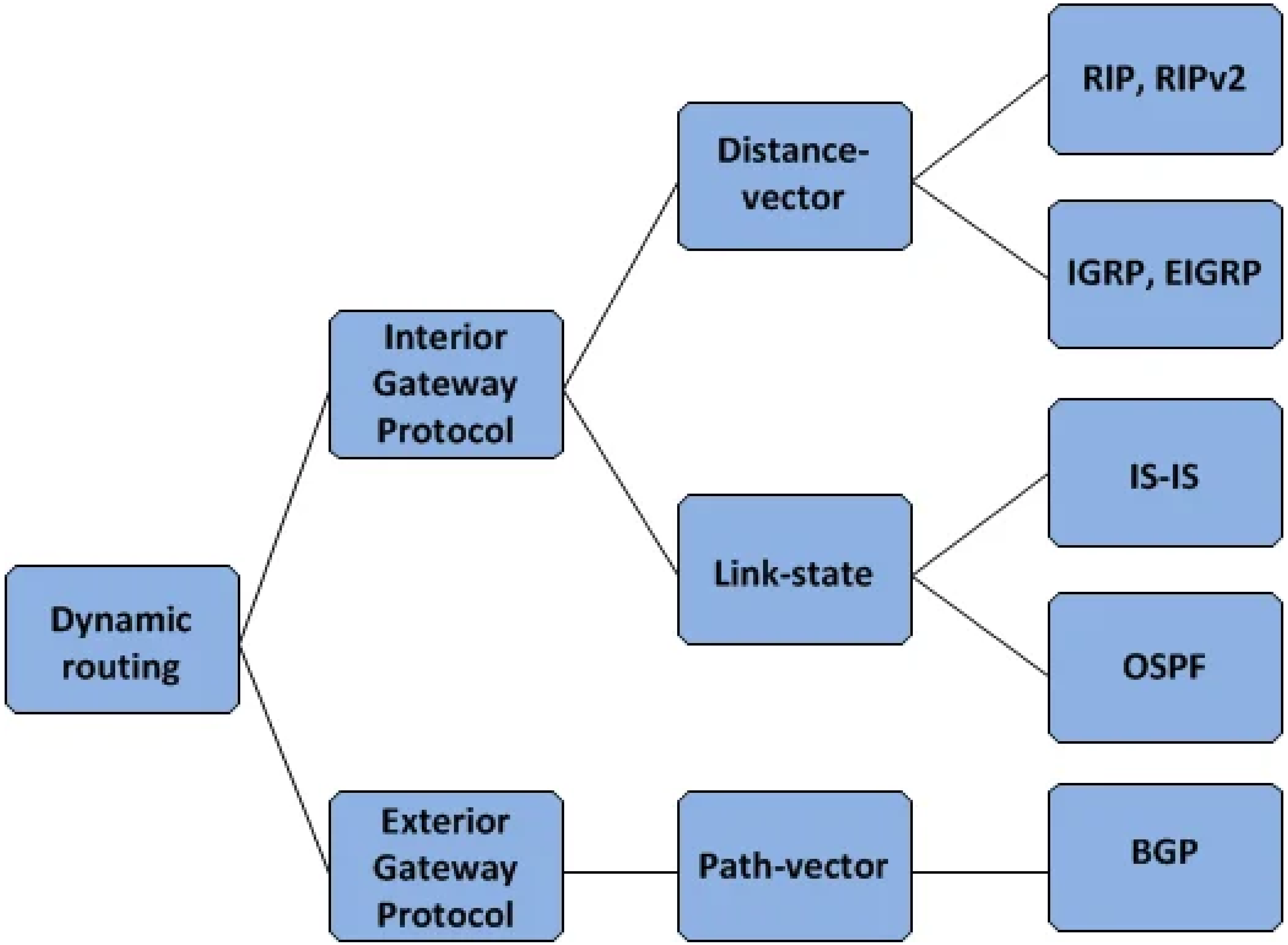
Inconvenient

Slow

Easy to make a mistake

Dynamic routing

Dynamic routing is a networking technique where routers automatically learn network topology and update routing tables in real-time, enabling adaptation to changes such as link failures or traffic congestion.

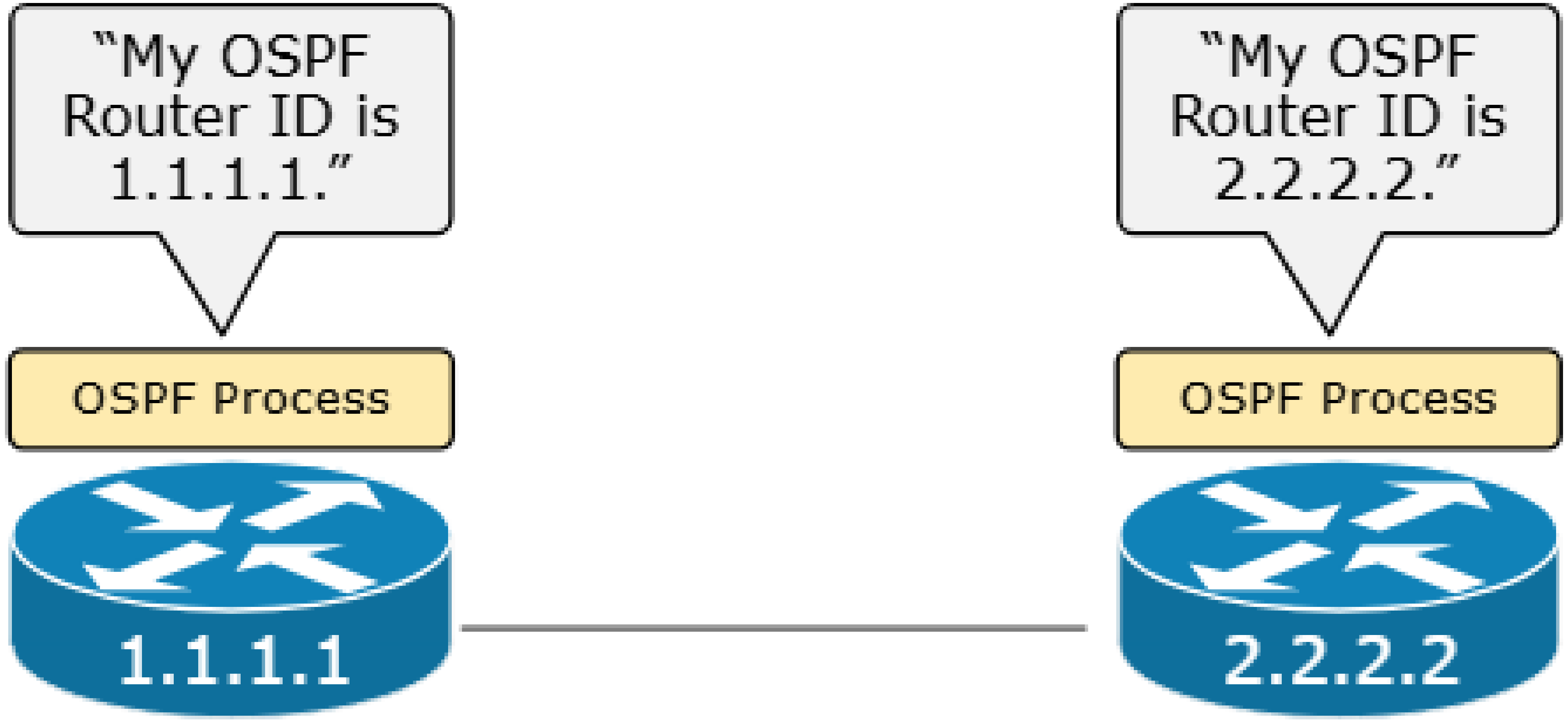


OSPF

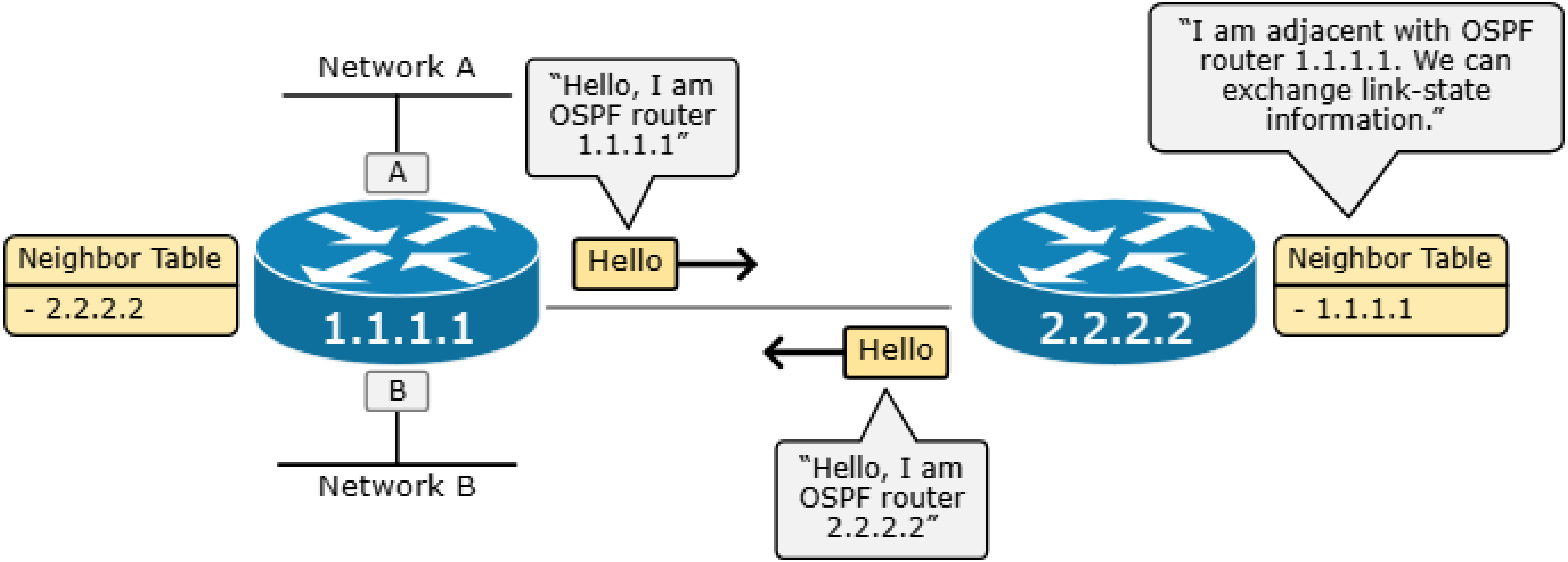
Open Shortest Path First (OSPF) is a link-state routing protocol used to find the best path between routers.

Each router knows the networks around itself → Routers share this information with each other using LSA (link state advertisements) → Routers form a LSBD (link state database) → Routers count the best path using LSDB and Dijkstra’s algorithm

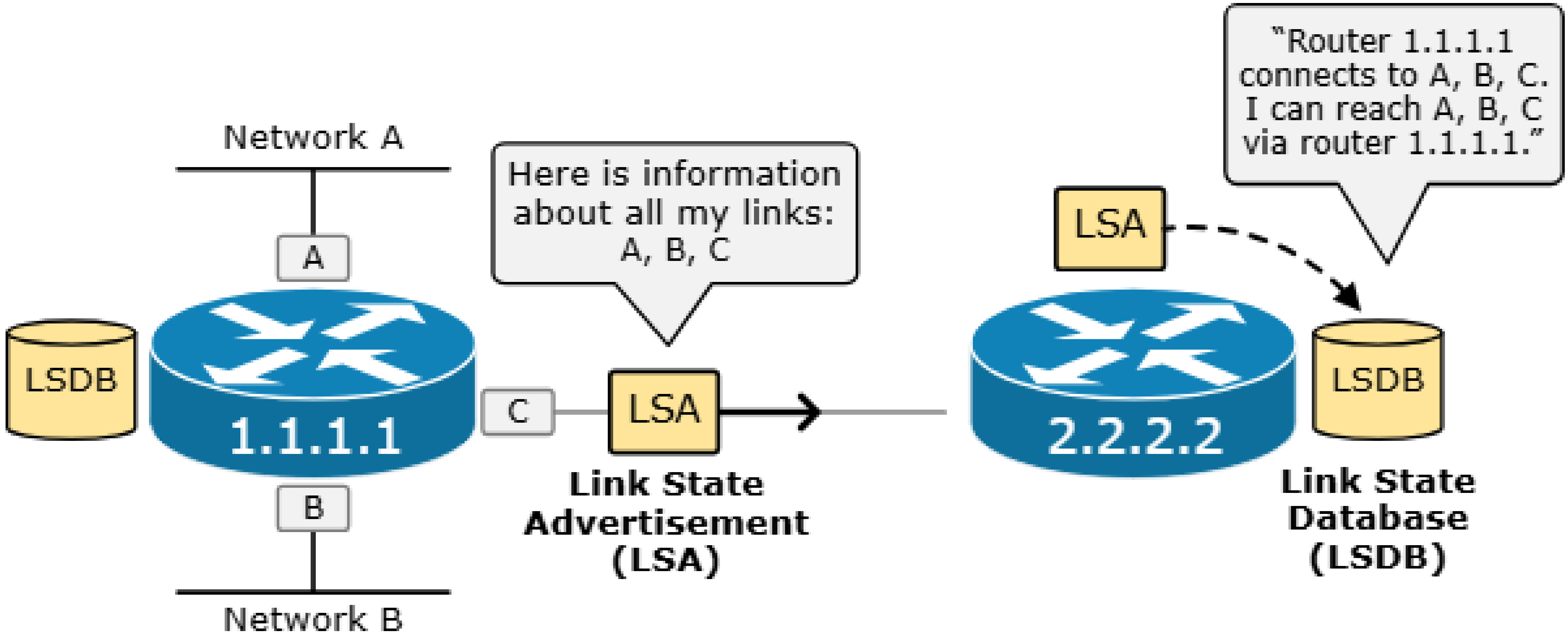
The Router ID

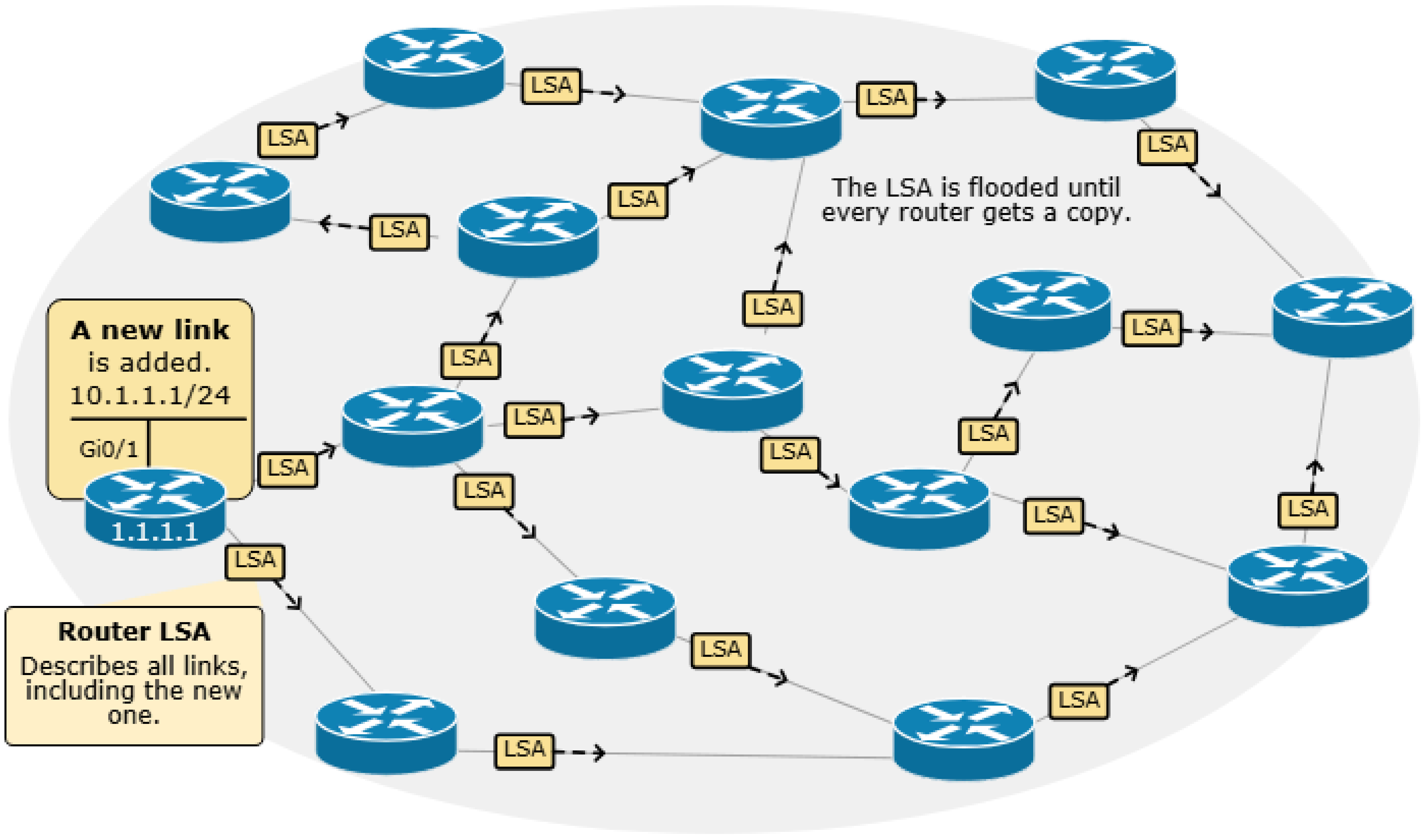


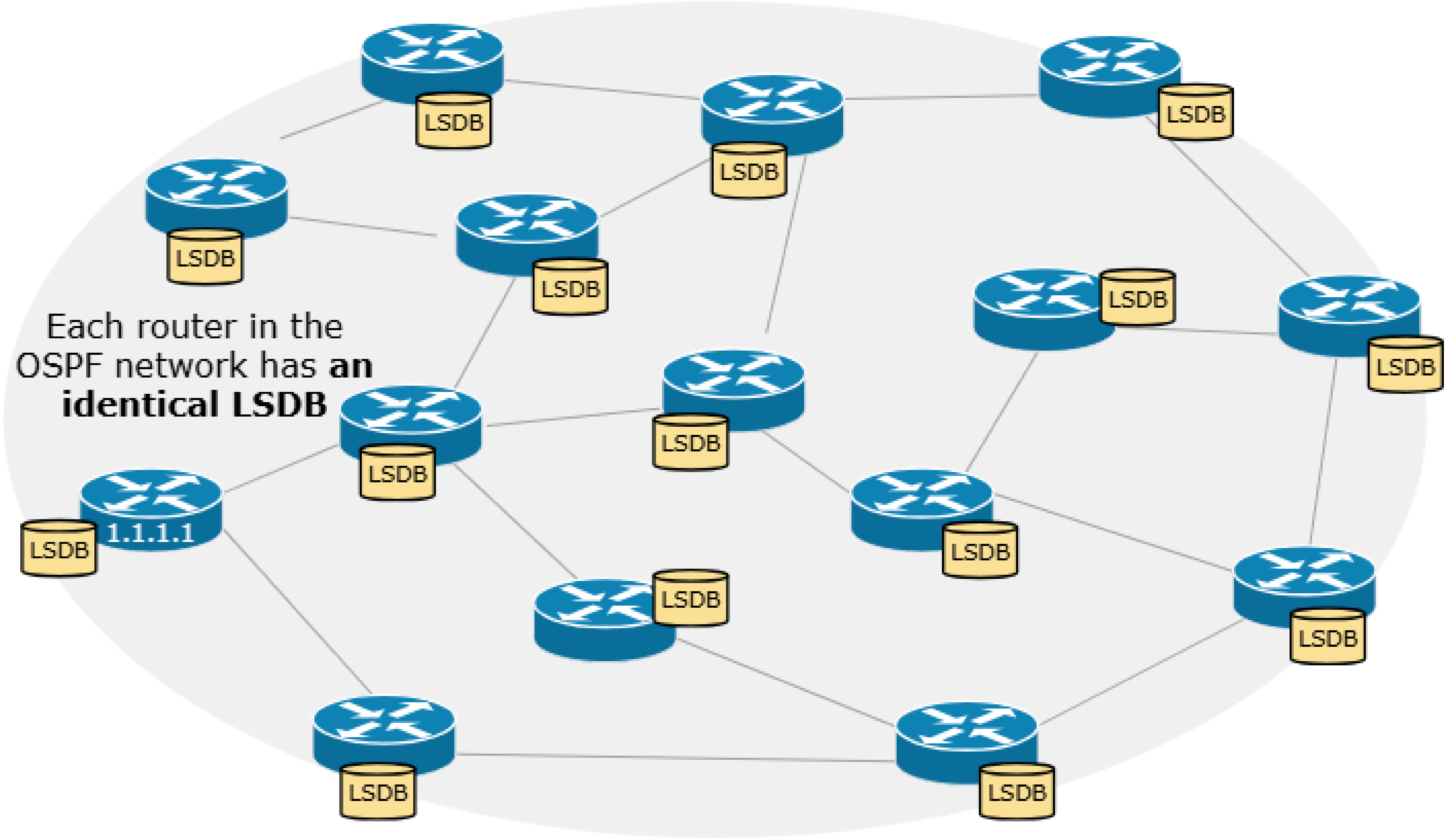
Becoming Neighbors



Exchanging Database Information

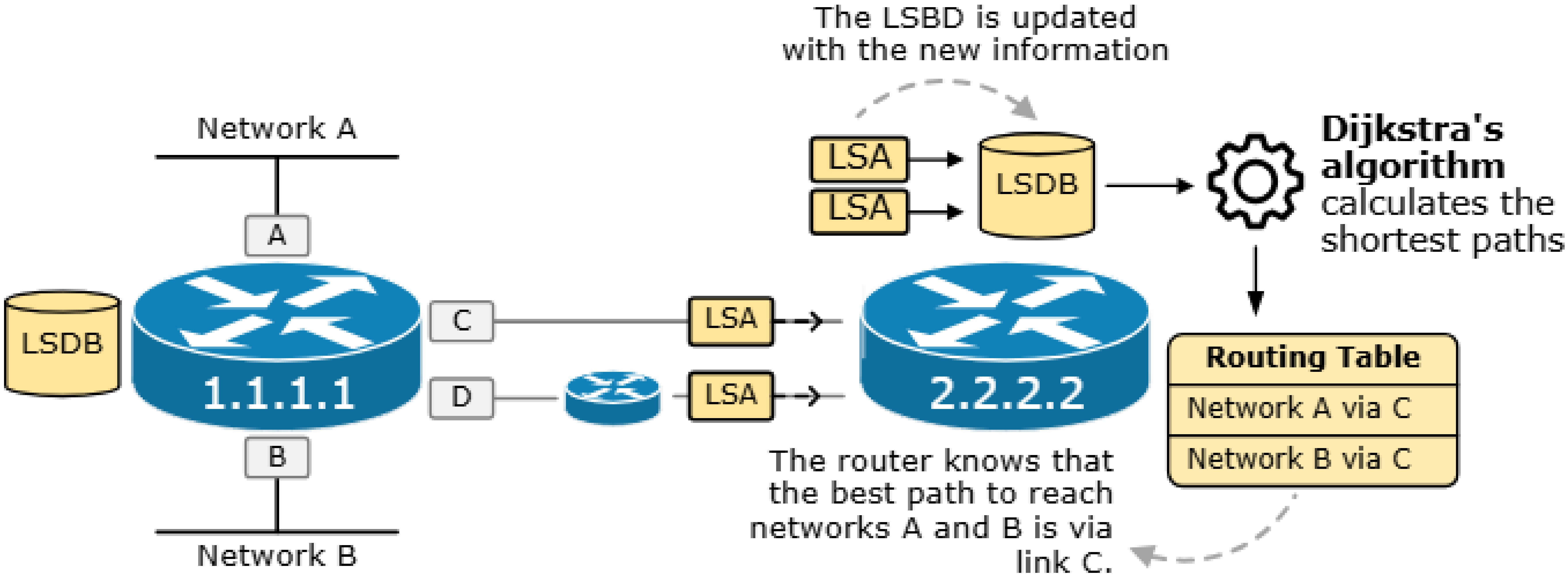



A new link is added.
10.1.1.1/24
Gi0/1
1.1.1.1
Router LSA
Describes all links, including the new one.
The LSA is flooded until every router gets a copy.
LSA

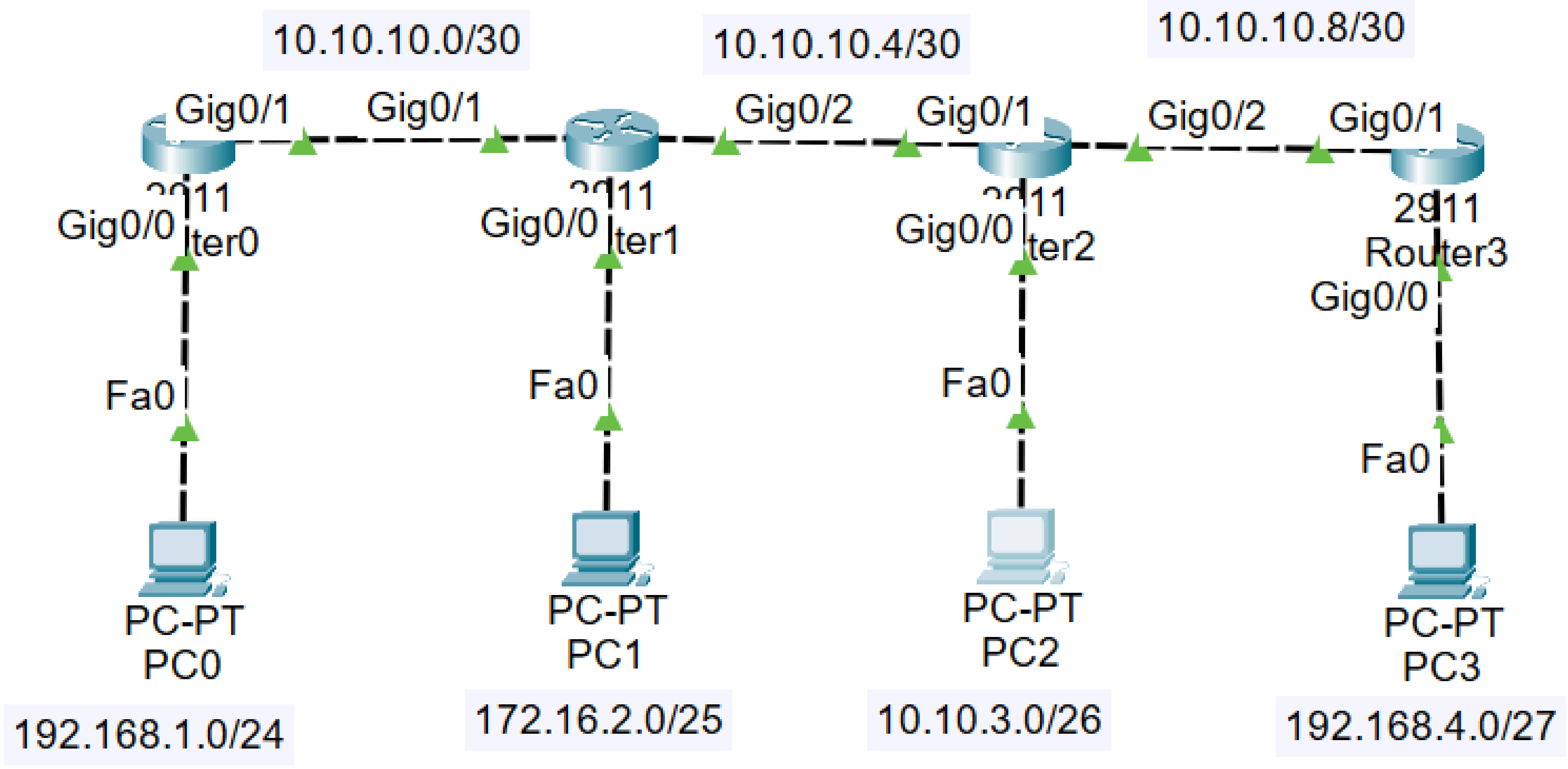


Each router in the OSPF network has **an identical LSDB**
1.1.1.1
LSDB

The SPF Algorithm



OSPF configuration



Step 1: Create OSPF process

```
Router(config)#router ospf 1
```

Step 2: Assign router-id

```
Router(config-router)#router-id 1.1.1.1
Router(config-router)#ex
```

Step 3: Configure directly connected networks

```
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
```

Network address

Wildcard mask

Wildcard mask

A **wildcard mask** is a 32-bit binary mask used in networking to determine which bits of an IP address to check or ignore when filtering traffic or defining networks. It is often called an **inverted subnet mask**, where 0 bits mean "must match" and 1 bits mean "ignore" (don't care).

255.255.255.0 → 0.0.0.255

255.255.224.0 → 0.0.31.255

Any questions?